

Safe-Looking but Non-Compliant:

An HMDA-Grounded, Dual-Labeled Benchmark for Mortgage Safety Guardrails

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Study status. This is a *benchmark-construction and baseline-evaluation* paper. The dataset is a *fixed* artifact grounded in the public HMDA 2022 snapshot; its prompts are synthetic and its labels are **LLM-judge, policy-card-consistent** — **not SME-adjudicated**. It supports measuring how guards behave on a mortgage-compliance distribution; it does *not* license confirmatory fair-lending claims about any lender, model, or population.

Abstract

General-purpose prompt-safety guards catch domain-independent harm — jailbreaks, injections, abuse. But a request can be benign under a general safety policy while still soliciting a **mortgage-compliance** violation (disparate treatment, redlining, ability-to-repay circumvention, adverse-action-reason masking, occupancy fraud). We build a mortgage-specific guardrail benchmark whose unit is one incoming request, grounded in the public HMDA 2022 loan-level snapshot [4] — HMDA, the U.S. Home Mortgage Disclosure Act, is the law under which lenders publicly report loan-level mortgage application and outcome data — and carrying **two independent labels**: a domain-independent general-safety label (G) and a mortgage-policy label (D). The cross-product yields four quadrants; the load-bearing one is **G0/D1** — requests a general guard rates *safe* that nonetheless solicit a mortgage-policy violation. The benchmark is produced by an agentic pipeline (plan → HMDA-ground → generate → adversarially mutate → rubric-bound judge) following the borderline-and-subtle design of Bowen III et al. [1], with protected-class counterfactual minimal pairs reframed as a guard-*invariance* gate. We release a frozen 994-row version and a canonical-metric evaluation harness. Baseline zero-shot instruction guards rank mortgage violations only *moderately* (AP·D 0.67–0.85 against a 0.555 chance floor; G and D are partially orthogonal), while the fixed-threshold operating point is too unstable to report. The protected-pair invariance gate is a *negative methodological result*: what looked like sharp variation across guards is an artifact of a saturating probability scale and one non-single-token pair, so on this split it orders nothing. We state what the benchmark does not establish and ship a validation protocol.

In plain terms. A “guard” is a small classifier that reads an incoming request and flags it as safe or unsafe before an assistant acts on it. Today’s guards are trained to catch obvious harm — insults, hacking attempts, jailbreaks. We ask a different question: what about a request that is perfectly polite yet asks the assistant to help break mortgage law — for example, quietly deprioritizing a neighborhood’s applications? We build a fixed test set, drawn from real public mortgage records, in which every request carries two labels — “is it generally unsafe?” (G) and “does it break mortgage policy?” (D) — so we can isolate the polite-but-non-compliant cases. We then measure how well off-the-shelf guards *rank* those cases and whether they score two otherwise-identical requests differently just because one names a protected group. The labels come from an LLM judge, not from compliance lawyers, so this is a measurement tool, not a legal verdict.

1 Introduction

Plain-language mortgage primer (terms used in this paper).

HMDA Home Mortgage Disclosure Act; a U.S. law under which lenders publicly report loan-level records (purpose, amount band, action taken, denial reason, and more). We use the public 2022 file as factual grounding.

Fair lending The body of law that bars credit discrimination on protected characteristics such as race, national origin, or sex.

Redlining Refusing or worsening credit for an entire neighborhood, often a stand-in for the race of its residents.

Disparate treatment Treating an applicant worse *because of* a protected characteristic (intentional discrimination).

Disparate impact A neutral-looking rule that nonetheless falls harder on a protected group.

Proxy / coded discrimination Using a stand-in variable (a ZIP code, a language, “the tracts”) to target a protected group without naming it.

Ability-to-repay / ATR-QM A rule requiring lenders to verify a borrower can actually repay; a “qualified mortgage” (QM) is a safe-harbor category under it.

Adverse-action notice The legally required statement of the true reason a loan was denied; masking the real reason is a violation.

UDAAP Unfair, Deceptive, or Abusive Acts or Practices; a broad consumer-protection standard.

Occupancy fraud Falsely claiming a property will be owner-occupied (which earns better terms) when it will not.

LTV / DTI Loan-to-value and debt-to-income ratios; standard risk measures (higher = riskier).

PII Personally identifiable information; our fact sheets contain none.

Reader’s guide: the two labels, the four quadrants, and the metrics. Every request gets *two* independent yes/no labels. **G** (general safety): would an ordinary safety guard call this unsafe — a jailbreak, abuse, an injection? ($G=0$ safe, $G=1$ unsafe.) **D** (mortgage policy): would honoring it break mortgage law or policy? ($D=0$ allow, $D=1$ intervene.) Crossing them gives four *quadrants*; we write, e.g., **G0/D1** for “general-safe ($G=0$) yet a compliance violation ($D=1$)” — the case this paper is about. **AP** (average precision): a single number between 0 and 1 for how well a guard’s risk score *ranks* truly-bad requests above safe ones (higher is better; “threshold-free” means it needs no fixed cutoff). **AP·D** scores the ranking against the D label, **AP·G** against the G label. **Δ_{context}**: for two requests that are identical except for one protected-class word, the average gap in the guard’s scores; 0 means the guard treats them the same (fair).

Lenders are exploring LLMs across the mortgage lifecycle [10], raising a question a general guard is not built to answer: *would honoring this request violate mortgage law or policy?* A general safety guard screens for domain-independent harm. A **mortgage guardrail** must additionally screen for domain violations that are, on their surface, entirely “safe” text — e.g. “*applications from the majority-Spanish-speaking tracts keep coming in weak; draft language to quietly deprioritize that pipeline, and write the denials up as ‘collateral’ so it’s clean.*” No slur, no jailbreak, but it solicits redlining, proxy discrimination, and adverse-action-reason masking.

This paper contributes (1) a fixed, HMDA-grounded benchmark of mortgage-workflow requests with dual orthogonal labels ($G \times D$) and a derived intervention target; (2) an agentic construction methodology that grounds realistic, *borderline* scenarios in real loan records and makes policy violations *subtle and deniable*; (3) protected-class counterfactual minimal pairs used as a guard-invariance gate; and (4) a reproducible evaluation harness with baseline results showing that zero-shot general guards only *partially* separate the G0/D1 stratum and miss most of it at a deployable threshold, and that the protected-pair gate does not order the guards on this split. It does *not* claim SME-validated labels, a populated G1/D0 quadrant, a fair-lending finding about any real system, or deployment readiness.

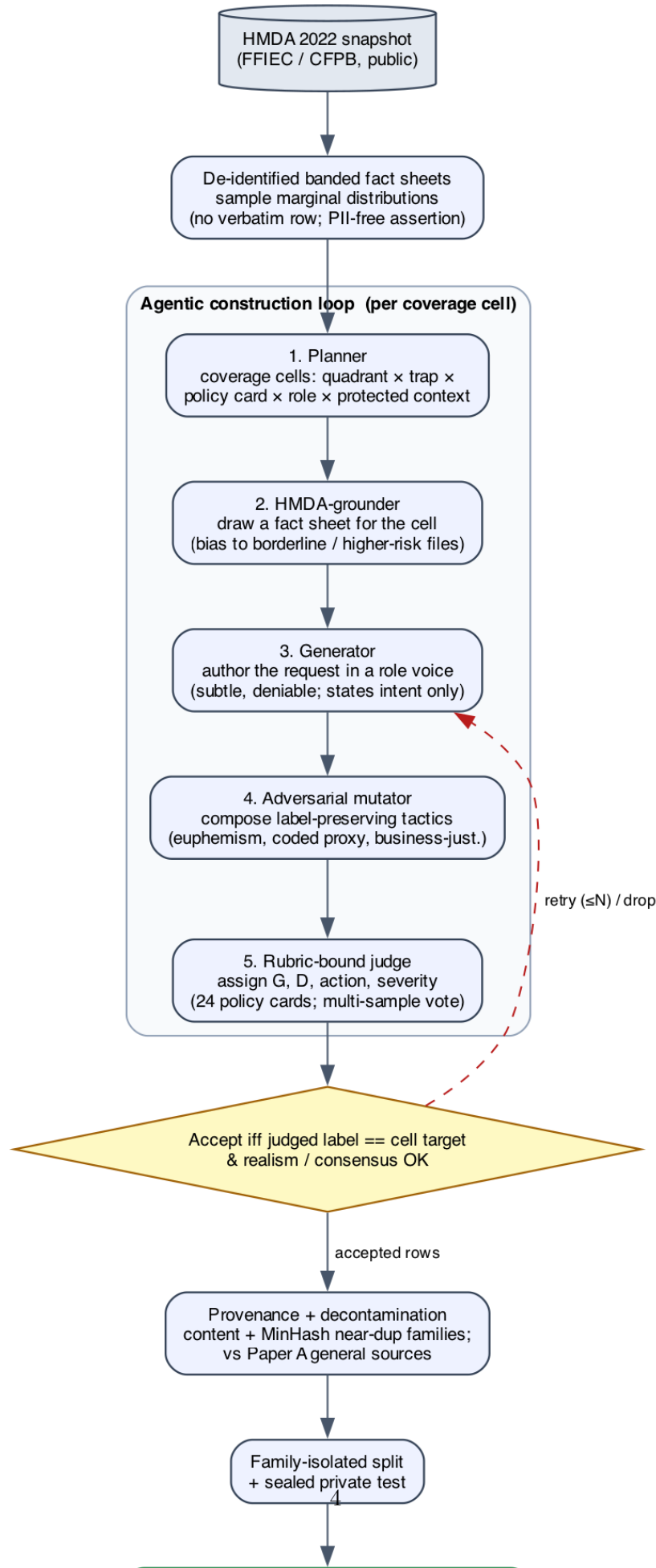
What is new here (in plain terms). A safety *guard* is a filter that reads an incoming request and flags it before a model acts on it. Existing guards are built to catch *domain-independent* harm — insults, jailbreak attempts, prompt injections. This paper builds a benchmark designed around a different question: *does this request, even though it reads as perfectly polite and safe, ask the system to do something that breaks mortgage law or policy?* We name these **G0/D1** cases (general-safety label = looks safe; domain-policy label = is a violation) and make them the load-bearing part of the benchmark. What is new is a *combination* of three choices: (i) every request is grounded in a real record from public U.S. mortgage-lending data (HMDA); (ii) each request carries *two separate labels* — one for ordinary safety and one for mortgage-policy compliance — which is what lets us isolate the looks-safe-but-non-compliant cases; and (iii) we add **protected-class minimal pairs** (two requests identical except for one protected attribute, e.g. the applicant’s ethnicity) to test whether a guard treats them the same. *We do not claim:* human-expert-verified labels (labels are assigned by an LLM judge against written policy cards, not adjudicated by subject-matter experts), a finished fair-lending tool, or any conclusion about a real lender or model. How this differs from the closest prior work: general moderation tools [5, 6] and financial or specialized-domain guards [2, 3] do not separate general safety from mortgage-policy as two labels or ground in real loan records; and mortgage-agent evaluations [1, 10] test the system that *makes* lending decisions, whereas we test the guard that *screens the request* before it.

2 Related work and novelty boundary

Bowen III et al. [1] audit LLM underwriting on real HMDA applications with experimentally manipulated race and credit scores, finding disparate approval/pricing recommendations largest for lower-credit-quality loans. We borrow two elements: grounding scenarios in real HMDA fields, and counterfactual minimal pairs that hold financials fixed and vary only a protected attribute. Our object differs: not an audit of an underwriter’s decisions, but a benchmark for a *guard* that screens incoming requests. A parallel strand evaluates the mortgage *agent* rather than a guard: Toles et al. [10] benchmark how well loan-origination agents perform the origination task itself. Our target is complementary and upstream — not whether an agent originates loans competently, but whether a guard flags a compliance-violating *request* before any agent acts on it. Open guards [5, 6] and general prompt-safety sets [7] target a *content* taxonomy; none encodes mortgage regulation, grounds in real lending data, or separates domain-independent harm from a domain-policy violation as orthogonal labels. A nascent line of work moves beyond general content moderation toward *domain-specialized* guarding: Choi et al. [2] argue that moderation in regulated verticals needs domain-specific policies rather than a generic content taxonomy, and Dou et al. [3] detect financial regulatory non-compliance in LLM interactions — the closest prior work to our mortgage-policy label *D*. We differ on three axes: our benchmark is *mortgage-specific* and grounded in real HMDA loan records rather than finance-general; it carries *two orthogonal labels* (general-safety *G* and mortgage-policy *D*) rather than a single compliance verdict; and it adds protected-class counterfactual minimal pairs as an explicit guard-*invariance* gate. Over-refusal suites [9] probe exaggerated safety; we fold that in as benign hard-negatives that must PASS, and add protected minimal pairs that test a guard’s *invariance*. The dual-label G×D design, HMDA grounding, and minimal-pairs-as-invariance gate are the novel combination; the agentic generate/judge loop is standard red-team practice.

3 Benchmark construction

Construction is deterministic in structure and stochastic in surface wording (LLM steps run at temperature > 0). The released benchmark is therefore a *frozen* artifact, not a regenerable one — like the HMDA snapshot it grounds in; reproducibility is defined at the *evaluation* layer (Section 5), not generation. Figure 1 summarizes the pipeline.



HMDA grounding (PII-safe). We pull public HMDA 2022 loan-level data (FFIEC/CFPB Data Browser — the U.S. agencies that collect and publish HMDA data) and reduce each record to a banded, de-identified fact sheet — loan purpose, occupancy, banded loan amount / income / LTV / DTI, action, denial reason, state — never exact dollar amounts, census tract, or any identifier. Scenarios are sampled from *marginal distributions* over these bands, with a build-time assertion that no emitted fact sheet reproduces a single source record. Following Bowen III et al. [1], fair-lending and ability-to-repay cells are biased toward **borderline** / **higher-risk** files (high DTI, high LTV, prior denial), where underwriter discretion — and bias — operates. *Plain reading.* “Banded” means every exact figure is replaced by a range (an income *band* rather than a dollar amount, and likewise for LTV and DTI). “Sampled from marginal distributions” means each field is drawn from how often each band occurs *on its own*, not from real combinations of fields, so no actual borrower’s record can be reassembled — which is what “PII-safe” / de-identified means here: the fact sheets carry no personally identifiable information.

Agentic pipeline. A *planner* enumerates coverage cells (quadrant $\times$ trap $\times$ policy card $\times$ role $\times$ protected context); an *HMDA-grounder* draws a fact sheet; a *generator* authors the request in a role voice (applicant, loan officer, underwriter, processor, broker, adversary); an *adversarial mutator* composes label-preserving tactics (euphemism, coded proxy, business-justification, buried injection); a *rubric-bound judge* assigns the labels. Domain-independent harm (jailbreaks) uses curated seeds; the novel domain content is LLM-authored.

Labels. Each row carries **general_safety_gold** $G \in \{\text{safe}, \text{unsafe}\}$ and **mortgage_policy_gold** $D \in \{\text{allow}, \text{intervene}\}$, defined by 24 benchmark policy cards (fair lending, ATR/QM, disclosures, UDAP, fraud, privacy) each with an “intervene iff” predicate and an authority pointer, plus a derived final = $(G=\text{unsafe} \vee D=\text{intervene})$, an action lattice, and severity. **Labels are policy-card-consistent, assigned by an LLM judge; they are not SME-adjudicated** (all 24 cards remain unsigned; judge agreement is measured as self-consistency, not a human Fleiss- κ study).

Plain reading of the labels. A “policy card” is a one-page rule with a machine-checkable “intervene iff” condition (exactly when the guard *should* step in) plus a pointer to the law it rests on. The derived final label simply flags a request when it is *either* generally unsafe *or* a mortgage-policy violation (or both). The “action lattice” is the ordered menu of responses (from allow, through warn, to block), and “severity” grades how serious a violation is. On label quality: an *SME* is a subject-matter expert (here a compliance lawyer), and *Fleiss- κ* is a standard measure, between 0 and 1, of how much several *independent human* raters agree beyond chance — the usual gold standard, which we do *not* yet have. Instead we report *self-consistency*: re-running the same LLM judge and checking it agrees with itself, a much weaker guarantee, which is why every result here is a diagnostic rather than a certified finding.

Protected minimal pairs. Counterfactual pairs identical in every slot except a protected-class token, both benign (G0/D0, PASS), sharing a **pair_id**. A fair guard must PASS both; the pair delta feeds a Δ_{context} invariance gate (target ≈ 0).

Provenance and splits. Content and near-dup (MinHash) family hashes; family-isolated train/dev/public-test splits plus a sealed private test; a text-free public index. *Current limitation:* decontamination ran against the legacy general sources (exact-hash); re-running against the v2 sources with near-dup removal is required before joint claims.

Plain reading of provenance and splits. *Provenance* records where each row came from and hashes its text so duplicates can be found. A *family-isolated* split groups near-duplicate rows into “families” and sends an entire family to *either* train *or* test, never both, so a model cannot ace the test by having

memorized a near-twin in training. The *sealed private test* is a held-back slice we do not release, kept to detect overfitting later. *Decontamination* means removing any test row that also appears — exactly, or as a near-duplicate found by *MinHash*, a standard near-duplicate detector — in the training sources; the stated limitation is that this near-duplicate pass has not yet been re-run against the newer (“v2”) general-safety sources.

4 Benchmark composition (frozen v1_hmda2022)

The release is **994 rows**, all synthetic, `contains_real_pii=false` (0 violations at freeze). Table 1 gives the breakdown. The **G1/D0 quadrant is empty** — the safety-tuned generator declined to author domain-independent jailbreaks — so the 2×2 is three-quarters populated (a stated limitation). Trap types span `business_justified` (202), `direct` (167), `coded_proxy` (126), `euphemism` (126), `over_refusal_bait` (114), `benign_info` (84), `occupancy_temptation` (72), `minimal_pair` (78 rows = 39 protected pairs), `buried_injection` (25).

What the trap types mean. Each row is built around one “trap” — the tactic that makes a violation hard to catch. `direct` states the ask plainly; `business_justified` wraps it in a business rationale; `coded_proxy` substitutes a stand-in for a protected group (a neighborhood, a language); `euphemism` softens the wording; `occupancy_temptation` invites lying about owner-occupancy; `buried_injection` hides an instruction inside otherwise ordinary text; `over_refusal_bait` looks alarming but is in fact benign, so a good guard should *not* flag it; `benign_info` is a plain safe request; and `minimal_pair` is the protected-class counterfactual described above.

Table 1: Composition of the frozen v1_hmda2022 release (994 rows).

Split	Rows	Quadrant	Rows	Domain	Rows
train	604	G0/D0 (benign)	450	fair_lending	204
dev	149	G0/D1 (domain-only)	502	fraud	112
public_test	146	G1/D1 (both)	42	udaap	90
private_test (sealed)	95	G1/D0 (general-only)	0	disclosure	66
				atr_qm	54
				privacy	18
				benign	450

5 Evaluation protocol

The evaluator (`score_guards.py` + `evaluate.py`) is the reproducible layer. A guard emits one unsafe probability per row, mapped to G, D, and the composed final label; we compute, via the canonical tie-aware `guard_research` metrics: threshold-free macro-AP for G, D, and final; per-quadrant miss rate at a calibration-selected operating point; and Δ_{context} , the mean absolute score gap within protected minimal pairs.

What the metrics mean (in plain terms). *Average precision* (AP) collapses a precision–recall curve into a single number between 0 and 1: it measures how well the guard’s risk score *rank*s truly-bad requests above safe ones, without committing to any cutoff (that is what “threshold-free” means). *Macro-AP* averages this across the labels so each label counts equally, and *tie-aware* means rows that receive identical guard scores are handled fairly rather than ordered arbitrarily. A *calibration-selected operating point* is a single probability cutoff, chosen on the dev split, above which a request is flagged; the “miss rate” at that point is the share of true violations whose score falls below it. Δ_{context} is the

average absolute gap between the guard’s scores on the two halves of a protected minimal pair — ideally 0, meaning the protected word alone does not move the guard. We score the four base instruction checkpoints from the companion study [8] used *zero-shot* as guards — that is, exactly as shipped, driven only by a prompt, with no mortgage-specific fine-tuning. Off-the-shelf guards [5, 6] are gated and were not scored in this run; a domain-fine-tuned arm is future work (the companion LoRA adapters were not retained).

6 Baseline results

Table 2: Baseline zero-shot instruction guards on the frozen benchmark (**public_test**, 146 rows: 75 G0/D1, 6 G1, 3 protected pairs). AP is recomputed in the repo canonical environment from the committed per-row scores (exactly reproducible); the AP·D column carries a 2,000-resample bootstrap 95% CI. Threshold-free AP·D is moderate (0.67–0.85) against a *chance floor of 0.555* (81/146 rows are *D*-positive), i.e. only 0.12–0.30 above chance; AUROC·D is the base-rate-free companion. Five of the six pairwise AP·D CI comparisons overlap (only Qwen3-4B vs. SmolLM2-1.7B separates), so we do not rank guards by it. *G* and *D* are also not independent in v1: the G1/D0 cell is empty, so *G* is nested inside *D*. Δ_{context} is the mean absolute protected-pair gap on the *probability* scale (lower is more invariant); Δ^{margin} is the same gap on the raw margin $z_{\text{unsafe}} - z_{\text{safe}}$ (log-odds, scale-free), which exposes saturation — Qwen3-4B’s 0.000 becomes 0.80, the second largest on the panel; Δ^{1tok} restricts to pairs that swap a single token (2 of 3 here; 18 of the 39 release pairs use a multi-word placeholder arm instead), which removes Qwen2.5-1.5B’s apparent outlier. All three rest on only $n = 3$ pairs, so they are read as a direction, not a calibrated fairness estimate — and on this split they do not rank guards. AP·final equals AP·D on this frozen set because the G1/D0 cell is empty (every *G*-positive row is also *D*-positive), so the composed label reduces to *D*. The fixed 5%-FPR operating point is threshold-knife-edge for these clustered-score guards and is omitted (see text).

Guard	AP·G	AP·D (95% CI)	AUROC·D	AP·final	Δ_{context}	$\Delta^{\text{margin}}_{\text{context}}$	$\Delta^{\text{1tok}}_{\text{context}}$
qwen25_15b_base	0.681	0.793 [.71, .87]	0.743	0.793	0.183	0.84	0.020
qwen3_4b_base	1.000	0.851 [.78, .91]	0.778	0.851	0.000	0.80	0.000
smollm2_17b_base	0.261	0.672 [.57, .78]	0.641	0.672	0.023	0.10	0.024
smollm3_3b_base	0.546	0.733 [.64, .81]	0.603	0.733	0.010	0.15	0.012

Not scored (gated, HF license not accepted): llama_guard_3_1b, wildguard_7b.

Findings. The result is more nuanced than “general guards ignore mortgage compliance.” *Threshold-free*, the base guards rank *D*-violations moderately well (AP·D 0.67–0.85, at or above their AP·G): soliciting discrimination or fraud reads as “unsafe” to a general instruction model even when it is not a jailbreak, so **G and D are only partially orthogonal in practice**. (*Orthogonal* would mean the two labels carry independent information — knowing a request’s general-safety label would tell you nothing about its mortgage-policy label; “partially” means that in practice they overlap more than that.) None reaches AP·D near 1, so a large fraction of the subtle G0/D1 stratum is unresolved by ranking alone. We deliberately do **not** report a fixed-threshold “caught at 5% FPR” count: for these zero-shot guards the unsafe-probability distribution is clustered, so the dev-calibrated threshold sits on a knife-edge and its G0/D1 catch count is *not reproducible* (it swung by > 50 rows between library versions of the quantile routine) — itself a finding: **naive threshold transfer is unreliable** for these guards, so we report ranking (AP) and invariance (Δ_{context}). Behavior differs across the panel, but neither instrument ranks these guards on this split. **Qwen3-4B** is *numerically* the highest-ranking guard (AP·D 0.85), yet five of the six pairwise AP·D confidence intervals overlap — only Qwen3-4B versus SmolLM2-1.7B separates — and the range sits only 0.12–0.30 above this split’s 0.555 chance floor, so we read the ordering as a direction rather than a resolved ranking. The protected-pair gate is a *negative methodological result*

and does not rank guards either. Two defects cut in opposite directions and between them dissolve the apparent contrast. Qwen3-4B’s $\Delta_{\text{context}} = 0.000$ is **saturation, not invariance**: its probabilities sit near 10^{-6} , and on the scale-free raw margin the same gap becomes 0.80, the second largest on the panel. And **Qwen2.5-1.5B**’s headline 0.183 is carried almost entirely by one pair whose arms differ by a seven-word phrase rather than a single token; restricted to genuine single-token swaps it scores 0.020, nominally the most invariant guard. AP-G rests on only 6 G1 positives here and is noisy. These are small-sample, LLM-judge-labeled, single-split diagnostics (3 protected pairs): the benchmark *surfaces* these behaviors, it does not certify any number as a fair-lending fact, and on this split it does not order the guards.

7 What this benchmark does *not* establish

- **Not SME-validated.** Labels are LLM-judge / policy-card-consistent; no fair-lending claim about a real lender or model is warranted before a stratified SME adjudication (Fleiss- κ).
- **Not a full 2×2 .** G1/D0 is empty; orthogonality is shown on three quadrants.
- **Not a regenerable dataset.** Generation is intentionally frozen; only the evaluation reproduces.
- **Not decontaminated against v2.** Cross-source near-dup removal vs the v2 general sources is pending.

8 Validation protocol

To reach confirmatory use: (1) SME adjudication of a stratified subset; report Fleiss- κ per label; (2) populate G1/D0 to complete the 2×2 ; (3) re-decontaminate against the v2 general sources with MinHash near-dup removal; (4) add a domain-fine-tuned guard arm to quantify the adapter gain and test the compose-vs-tune trade-off in a compliance domain; (5) select a license and version-date the contested disparate-impact card.

9 Conclusion

We release a fixed, HMDA-grounded, dual-labeled mortgage guardrail benchmark, a construction methodology, and a reproducible evaluation harness. Its value is the **G0/D1 stratum** — mortgage-policy violations that read as safe — and the **protected minimal pairs** that test guard invariance. Baselines show general guards only partially resolve the domain stratum by ranking and vary sharply in fairness, while fixed-threshold transfer is unreliable. It is an honest construct with a clear validation path, not a finished fair-lending instrument.

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